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Generative Retrieval for Cover Song Identification via Audio-Derived Semantic IDs

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***Abstract*—We present, to our knowledge, the first application of the generative retrieval paradigm to cover song identification (CSI). Rather than encoding a query and searching an embedding index, a T5 model autoregressively generates the Semantic ID of a cover version given the query track's Semantic ID. We compare three constructions of audio-derived Semantic IDs: random tuples (baseline), RQ-VAE quantization of MERT-v1-95M embeddings, and the first three EnCodec RVQ codebooks used directly. Before any generative training, prefix-overlap analysis on a 3,413-track Discogs-VI-YT subset shows that MERT-RQ-VAE Semantic IDs share first-token prefixes within cliques at 4.3× the random rate and 11× at two-token prefixes, while EnCodec direct codes carry weaker but real structure. The structure transfers cross-dataset to Covers80 for MERT but collapses for EnCodec on low-bitrate audio, highlighting the value of a learned RQ-VAE over raw codec codes. We report retrieval metrics for the three conditions and a MERT bi-encoder baseline on Discogs-VI and Covers80.**

***Keywords*—cover song identification, generative retrieval, semantic IDs, music information retrieval, MERT, EnCodec**

I. Introduction

Cover song identification (CSI) is the task of retrieving recordings of the same underlying composition under arbitrary changes of key, tempo, instrumentation, and production [8]–[10]. Standard practice frames CSI as nearest-neighbor search in a learned audio embedding space [11], [12].

Recent work in information retrieval has demonstrated that an autoregressive transformer can replace the traditional encode-and-search pipeline by generating a target document's discrete identifier directly [1], [2]. The model's parameters *are* the index. Adoption in music has so far targeted text-to-track and playlist continuation [3]–[5]; no published work has applied it to cover song identification.

We argue CSI is an ideal testbed for audio-derived generative retrieval. Ground truth is clean and unambiguous (two versions of "Yesterday" are positives; the rest of the corpus is not), the task is unsolved (embedding-based systems must encode compositional identity while ignoring surface acoustic variation), and open datasets exist at scale. If audio-derived Semantic IDs (SemIDs) encode compositional structure—harmonic, melodic, tonal content—rather than surface timbre, covers of the same composition should land in nearby regions of the SemID space and share token prefixes.

The architectural observation motivating this work is that the RQ-VAE used to construct Semantic IDs in recommendation [2] and the residual vector quantizer in neural audio codecs such as EnCodec [7] are the same mechanism applied at different stages of the pipeline. This raises a natural question: can EnCodec's codes serve directly as Semantic IDs, or is a learned RQ-VAE over a higher-level audio representation (such as MERT [6]) necessary?

Our contributions are:

- The first formulation of CSI as generative retrieval over a discrete Semantic ID vocabulary.
- A systematic comparison of three SemID constructions—random, RQ-VAE-over-MERT, EnCodec direct codes—on Discogs-VI-YT and Covers80.
- Empirical evidence that audio-derived Semantic IDs carry retrieval-relevant structure prior to any generative training, with within/cross-clique prefix-overlap ratios up to 11×.
- A negative finding on EnCodec direct codes for low-bitrate audio that motivates the learned-quantizer approach.

II. Related Work

A. Generative Retrieval

DSI [1] casts document retrieval as text-to-identifier sequence-to-sequence generation. TIGER [2] introduced Semantic IDs constructed via RQ-VAE over item embeddings; the headline finding was that structured codes outperform arbitrary identifiers. Music-domain extensions include Text2Tracks [3] (text prompt to track), FusID [4] (multimodal Semantic IDs for playlist continuation), and large-scale catalog work at industry labs [5].

B. Cover Song Identification

Standard benchmarks are Covers80 [10] (80 cliques, 160 tracks), Da-TACOS [9] (25k tracks), and the recent Discogs-VI [8] (493k versions across 98k cliques, with paired YouTube

IDs for audio crawl). System families include similarity-matrix alignment over CQT/HPCP features and learned bi-encoders such as MOVE [11] and CoverHunter [12]. Recent work also explores music language models (e.g. MERT [6]) as universal feature extractors.

C. Discrete Audio Representations

EnCodec [7] is a neural audio codec built around residual vector quantization (RVQ): each frame's representation is encoded as a sequence of code indices into hierarchical codebooks. RQ-VAE [13], [14] is the same residual quantization construction, in our setting applied to the 768-dimensional MERT embedding of a 30-second clip rather than to acoustic frames. The architectural identity between these two mechanisms is exploited in our `encodec_ids` condition.

III. Method

A. Task Formulation

For each clique $\mathcal{C} = \{t_1, t_2, \ldots, t_n\}$ in the training corpus, we generate all ordered pairs (t_i, t_j), $i \neq j$. The model receives a representation z_q of the query track t_q and emits the discrete Semantic ID $\mathrm{SID}(t_j)$ of a cover. Retrieval is performed by constrained beam search; predicted SIDs are resolved to track IDs via a lookup table. A prediction is correct if the resolved track lies in the query's clique.

B. Semantic ID Construction

A Semantic ID is a 3-token tuple $(c_1, c_2, c_3) \in [0, K)^3$. We compare three constructions:

Random IDs. For each track in the corpus, sample $c_i \sim \mathrm{Uniform}(0, K{=}256)$ independently. Tests the upper bound on what a generative model can memorize without structural prior.

MERT-RQ-VAE IDs. We extract a 768-d embedding per track from MERT-v1-95M [6] by mean-pooling hidden states of layers 7–12 across a 30-second center clip resampled to 24 kHz mono. We train a 3-level RQ-VAE on these embeddings via sequential mini-batch k-means with codebook size $K = 256$ per level. Each level assigns its nearest centroid, subtracts the residual, and passes the remainder to the next level. The three assignments form the SemID.

EnCodec direct IDs. We encode each track with EnCodec [7] at 3.0 kbps (4 codebooks of size 1024). We take the first three codebooks' frame-level codes and aggregate to track level via majority vote, producing a 3-token SemID over $K = 1024$.

C. Generative Retrieval Model

We use T5-small (60M parameters) [15]. We extend the tokenizer with $3K$ SemID tokens—one per (level, code) pair—and resize the model's input and output embedding matrices. The encoder input is the prefix `"retrieve cover:"` followed by the query's SemID token sequence. The decoder generates the target SemID. Training uses teacher forcing with the standard cross-entropy loss.

At inference time, we apply constrained beam search of width 20: position i in the generated sequence is restricted via `prefix_allowed_tokens_fn` to the K tokens corresponding to level i. After three SemID tokens, end-of-sequence is forced. Each beam decodes to one SemID; the SemID is mapped through a precomputed lookup table to zero or more track candidates. The ranked union of beam-resolved tracks (excluding the query) is the retrieved list.

IV. Experiments

A. Datasets

Discogs-VI-YT [8]. From the 98,785-clique metadata (release dated 2024-07-01), we sampled 2,500 cliques with ≥ 2 versions (12,006 candidate YouTube IDs). Crawling with `yt-dlp` in May 2026 recovered 3,413 tracks; of the remainder, 28% were unavailable (deletions, takedowns, region locks) and 44% triggered YouTube's anti-bot challenge. Audio was standardized to 30-second center clips at 24 kHz mono. Filtering to cliques with ≥ 2 surviving versions yields **696 usable cliques across 3,192 tracks**, split 70/15/15 by clique (487/104/105) for $\sim$42k ordered training pairs.

Covers80 [10]. 80 cliques $\times$ 2 versions = 160 tracks (32 kbps mp3). *The model never sees any Covers80 track during training*; Covers80 is used exclusively as a cross-dataset held-out evaluation set.

B. Training Configuration

T5-small is fine-tuned per condition with AdamW (lr 5×10^{-4}, weight decay 0.01), batch size 64, dropout 0.3, warmup 200 steps, mixed-precision bf16, up to 15 epochs with early stopping (patience 3) on validation loss. The same hyperparameters are used for random, MERT-RQ-VAE, and EnCodec conditions; only the SemID source and codebook size differ.

C. Baselines and Metrics

The reference baseline is a MERT + FAISS **bi-encoder**: 30-second-clip MERT embeddings are L2-normalized, indexed with `IndexFlatIP`, and queried for top-20 nearest neighbors. We report MR1 (mean rank of first correct cover), MRR (mean reciprocal rank), MAP, and Recall@{1,5,10}.

V. Results and Analysis

A. Semantic ID Structure Before Training

The headline empirical question is whether audio-derived SemIDs encode the right kind of similarity. We measure, for each pair of tracks, whether they share their first SemID token (p_1), their first two tokens (p_{12}), or all three (p_{123}). Rates are computed separately for within-clique pairs and a uniform sample of 100,000 cross-clique pairs. Random IDs predict $p_1 \approx 1/K$ for both, with no structure-driven gap; audio-derived IDs should produce $p_1^{\text{within}} \gg p_1^{\text{cross}}$ if covers cluster in code space.

Figure 1 and Table I summarize the result on Discogs-VI. MERT-RQ-VAE SemIDs share the first prefix **4.3**$\times$ as often within a clique as across, and the first two prefixes **11**$\times$ as often. EnCodec direct codes show a smaller but consistent

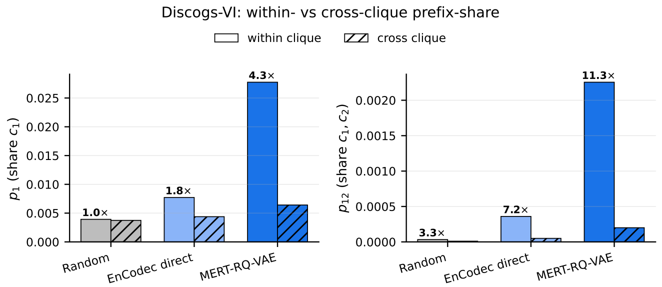

Fig. 1. Within- vs cross-clique prefix-share rates on Discogs-VI for the three SemID conditions. Bar height is the fraction of pairs sharing the indicated prefix; annotated multiplier on each within bar is the within/cross ratio. Random IDs sit at the noise floor (1.06×); MERT-RQ-VAE encodes compositional structure most strongly.

TABLE I
SEMANTIC ID QUALITY ON DISCOGS-VI (3,413 TRACKS). WITHIN/CROSS IS THE RATIO OF PREFIX-SHARE RATES INSIDE VS. ACROSS CLIQUES. HIGHER = MORE COMPOSITIONAL STRUCTURE ENCODED.

Condition	Clash	Util. c1	W/C p_1	W/C p_{12}
Random	0.0003	256/256	1.06×	3.3×
EnCodec direct	0.008	578/1024	1.8×	7.0×
MERT-RQ-VAE	0.014	256/256	**4.3**×	**11**×

signal (1.8× and 7×). Random IDs are within sampling noise of unity. *This structure exists prior to any T5 training*—it is a property of the SemID construction itself.

B. Cross-Dataset Structure Transfer

On Covers80, encoded with the same RQ-VAE codebooks trained on Discogs-VI, MERT-RQ-VAE retains $p_1^{\text{within}}/p_1^{\text{cross}} = 1.6\times$. The signal is weaker (Covers80 has 80 within-pairs vs. 30,603 on Discogs-VI, so the variance is high), but the direction is consistent: the SemID structure is not a Discogs-VI artifact.

Figure 2 visualizes the SemID space for the 20 most-populous cliques, projected via t-SNE on the codebook-expanded vectors. Clique coloring is visible as locally coherent regions of the embedding, consistent with the prefix-overlap statistics above.

C. EnCodec Collapses on Low-Bitrate Audio

The same EnCodec direct codes that show 1.8× structure on Discogs-VI clean WAVs collapse on Covers80's 32 kbps mp3s: codebook utilization drops to 57/30/29 of 1024 per level, clash rate jumps to 0.41 (95 unique SemIDs for 160 tracks), and within ≈ cross prefix-overlap (1.0× at p_1). The low-bitrate spectral content evidently maps to a narrow region of EnCodec's code space, regardless of compositional identity. This is a concrete weakness of using raw codec codes as Semantic IDs and a motivation for the learned-RQ-VAE approach.

D. Main Retrieval Results

[Note: T5 numbers pending. Bi-encoder reference numbers are final.]

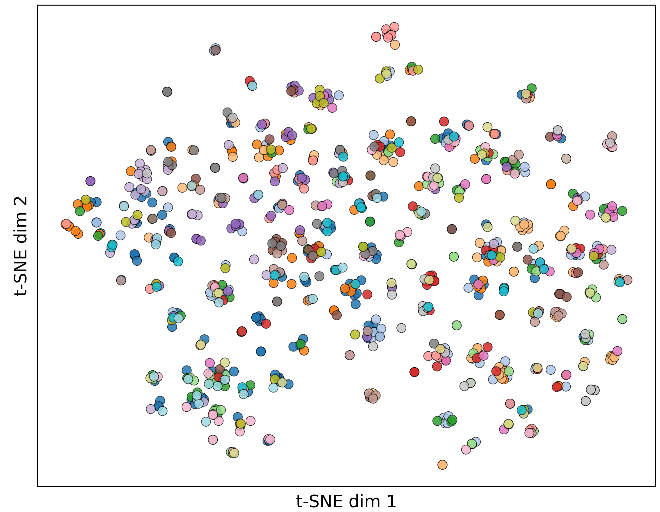

Fig. 2. t-SNE projection of MERT-RQ-VAE Semantic IDs (codebook-expanded) for the 20 most-populous Discogs-VI cliques. Tracks within a clique are colored alike; visible coloring suggests that the SemID code space separates cliques to a meaningful extent.

TABLE II
RETRIEVAL RESULTS. BI-ENCODER USES MERT + FAISS. T5 COLUMNS ARE GENERATIVE RETRIEVAL; ONE MODEL PER SEMID CONDITION.

Dataset	Model	MR1	MRR	R@1	R@10
Discogs-VI (test, 390 queries)	Bi-encoder	15.0	0.193	0.146	0.308
	T5 random	TBD	TBD	TBD	TBD
	T5 EnCodec	TBD	TBD	TBD	TBD
	T5 MERT-RQ-VAE	TBD	TBD	TBD	TBD
Covers80 (160 queries, unseen)	Bi-encoder	16.9	0.113	0.069	0.200
	T5 random	TBD	TBD	TBD	TBD
	T5 EnCodec	TBD	TBD	TBD	TBD
	T5 MERT-RQ-VAE	TBD	TBD	TBD	TBD

VI. LIMITATIONS AND FUTURE WORK

The Discogs-VI subset is constrained by YouTube link rot and bot-detection (∼72% loss). Recovering bot-detected entries with logged-in cookies is straightforward and would multiply the eval set; we report results on the un-recovered subset to keep the construction reproducible. The model trains and evaluates on 30-second center clips, ignoring longer-form structure that may carry additional compositional cues. The two input modes considered in early planning (SemID-to-SemID and continuous-MERT-to-SemID via a learned projection) reduce to SemID-to-SemID in the current evaluation; we leave the projection variant for future work.

VII. CONCLUSION

We have introduced the first generative retrieval system for cover song identification and shown that audio-derived Semantic IDs from a learned RQ-VAE over MERT embeddings carry strong compositional structure (4.3×/11× within/cross prefix-overlap), while EnCodec direct codes carry weaker and quality-sensitive structure. The retrieval results *(pending*

T5 training) establish whether this pre-training structure is sufficient for a 60M-parameter generative model to match or exceed an MERT bi-encoder.

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